# Business Intelligence – Data Mining


**Prepared By:**


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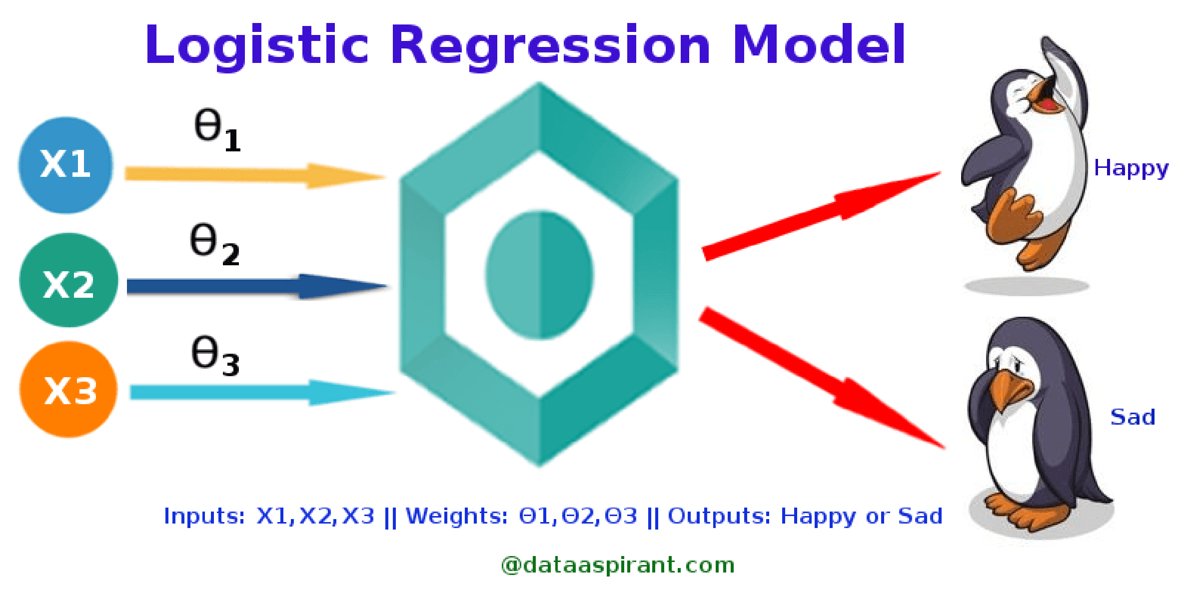
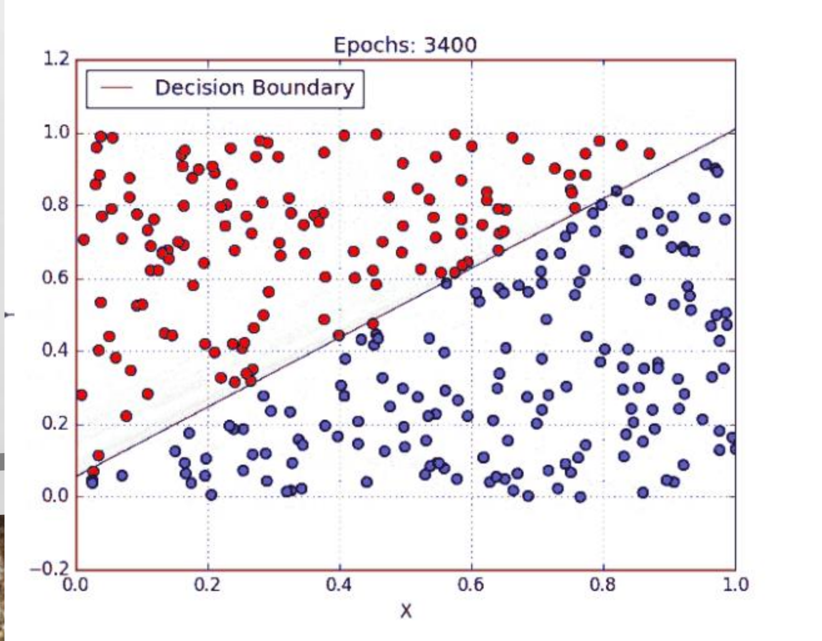
# Introduction to Logistic Regression

Logistic regression is a classification algorithm used to assign observations to a discrete set of classes. Some of the examples of classification problems are Email spam or not spam, Online transactions Fraud or not Fraud, Tumor Malignant or Benign. Logistic regression transforms its output using the logistic sigmoid function to return a probability value.

**What are the types of logistic regression**

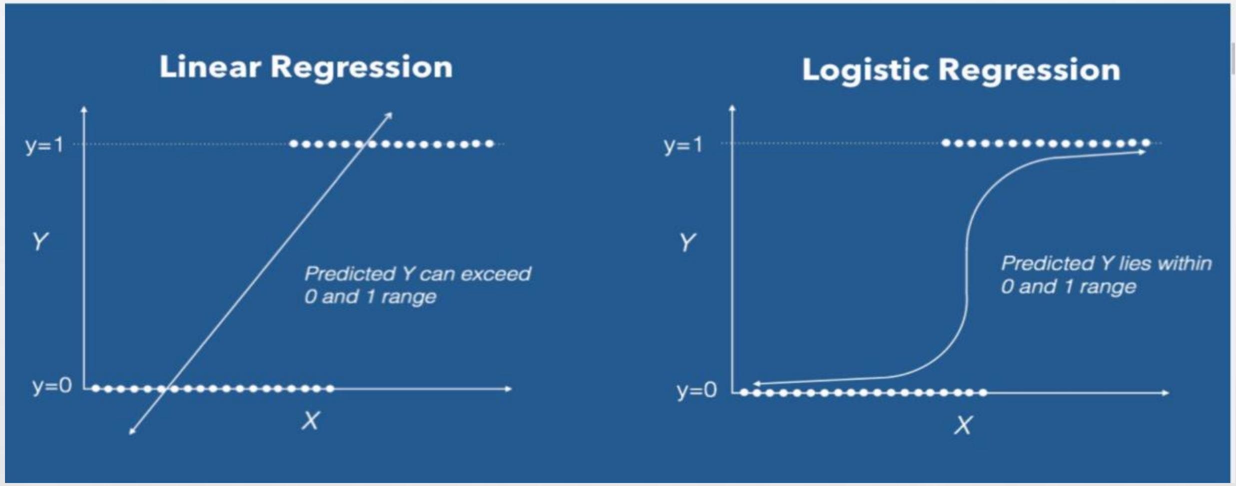
- Binary (eg. Tumor Malignant or Benign)
- Multi-linear functions failsClass (eg. Cats, dogs or Sheep's)

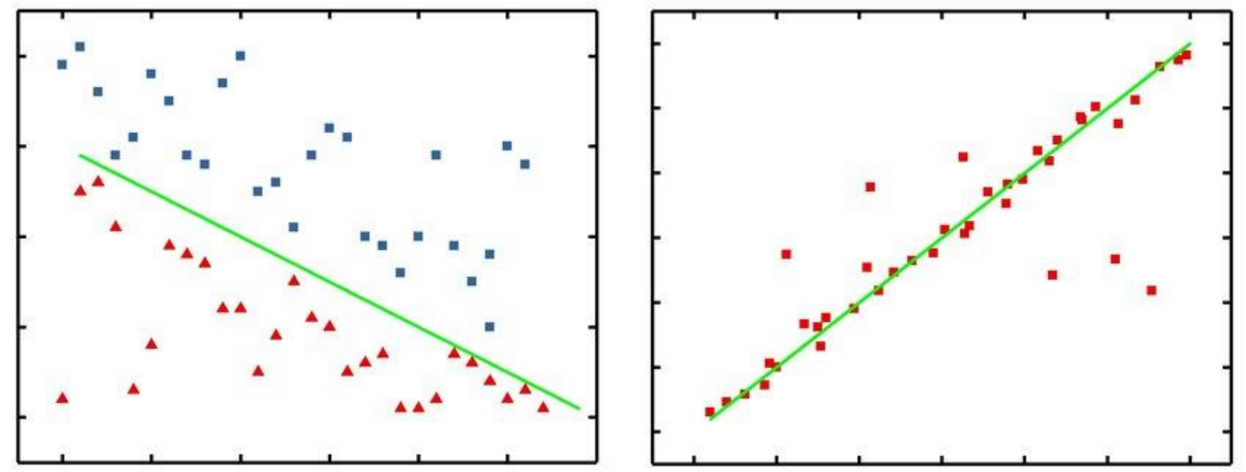


# Logistic Regression

Logistic Regression is a Machine Learning algorithm which is used for the classification problems, it is a predictive analysis algorithm and based on the concept of probability.



(a) Logistic Regression

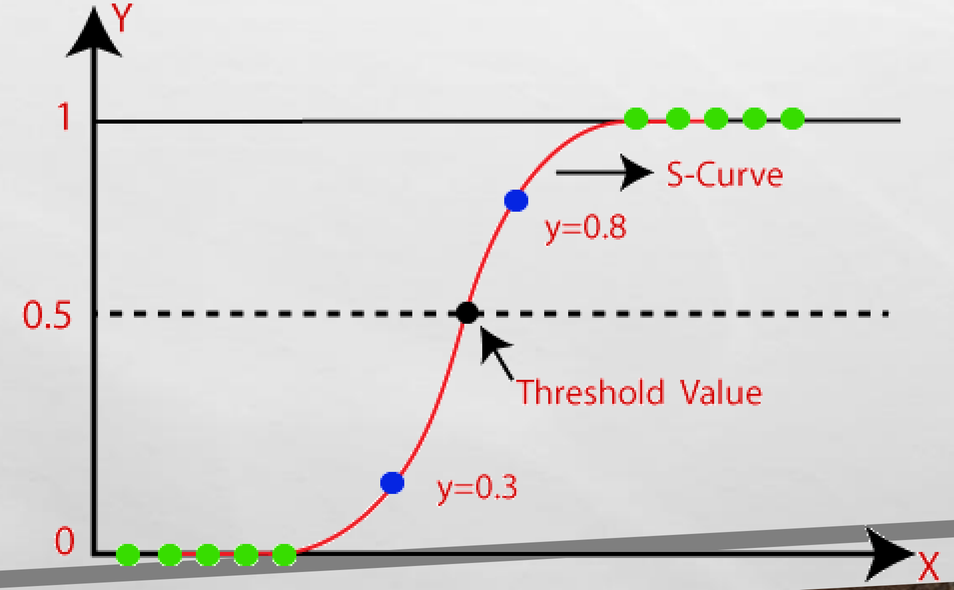
(b) Linear Regression

# Logistic Regression

- Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables.
- Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, **it gives the probabilistic values which lie between 0 and 1**.
- Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas **Logistic regression is used for solving the classification problems**.

# Logistic Regression

•In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).
•The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not based on its weight, etc.
•Logistic Regression is a significant machine learning algorithm because it has the ability to provide probabilities and classify new data using continuous and discrete datasets.
•Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification. The below image is showing the logistic function:



# Logistic Function (Sigmoid Function):

- The sigmoid function is a mathematical function used to map the predicted values to probabilities.
- It maps any real value into another value within a range of 0 and 1.
- The value of the logistic regression must be between 0 and 1, which cannot go beyond this limit, so it forms a curve like the "S" form. The S-form curve is called the Sigmoid function or the logistic function.
- In logistic regression, we use the concept of the threshold value, which defines the probability of either 0 or 1. Such as values above the threshold value tends to 1, and a value below the threshold values tends to 0.

# Assumptions for Logistic Regression:

- The dependent variable must be categorical in nature.
- The independent variable should not have multi-collinearity.

# Type of Logistic Regression:

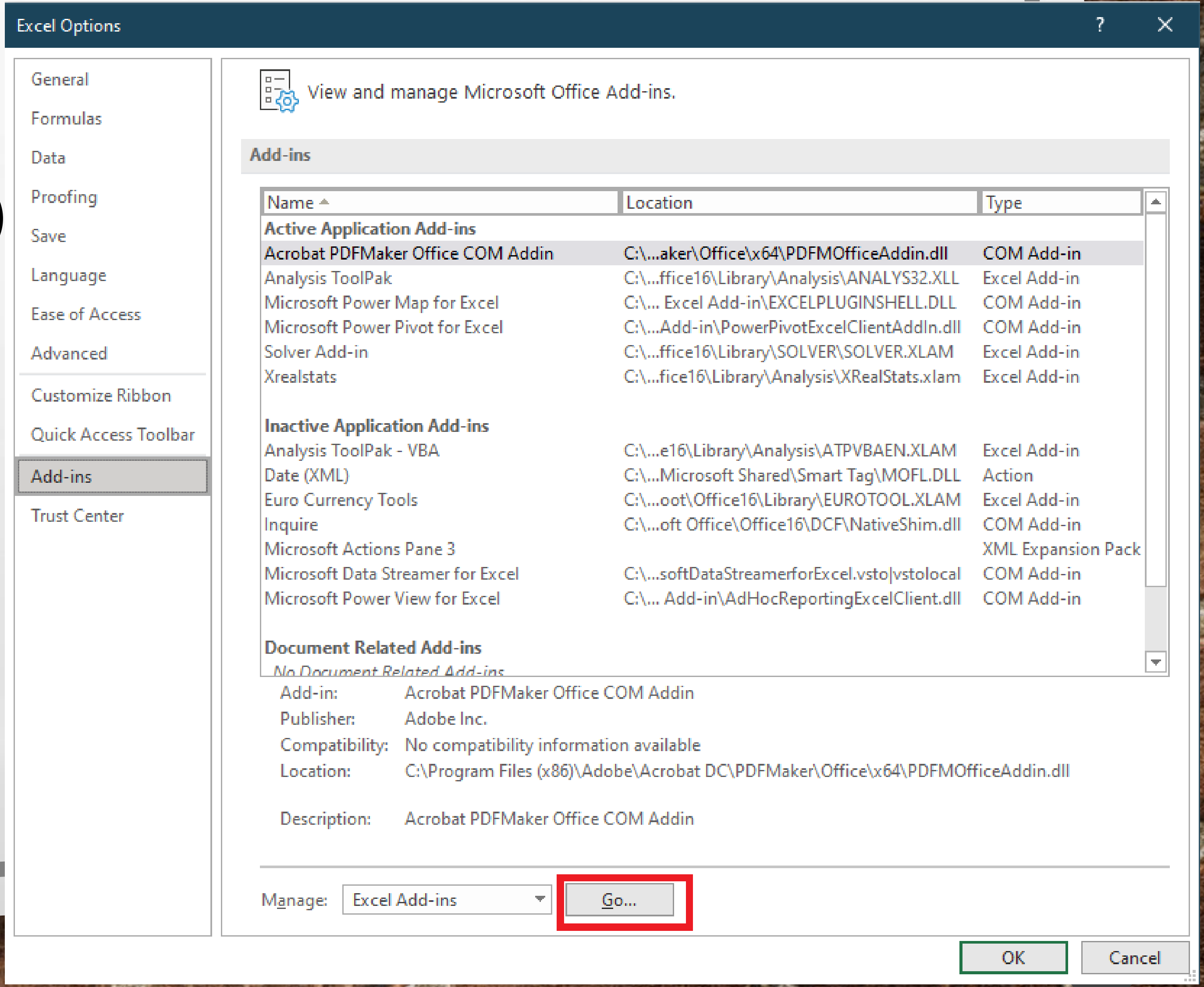
On the basis of the categories, Logistic Regression can be classified into three types:

- **Binomial:** In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
- **Multinomial:** In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as "cat", "dogs", or "sheep"
- **Ordinal:** In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as "low", "Medium", or "High".

# Steps for Using Logistic Regression- Using Real Statistics

**Installing – Real Statistics**

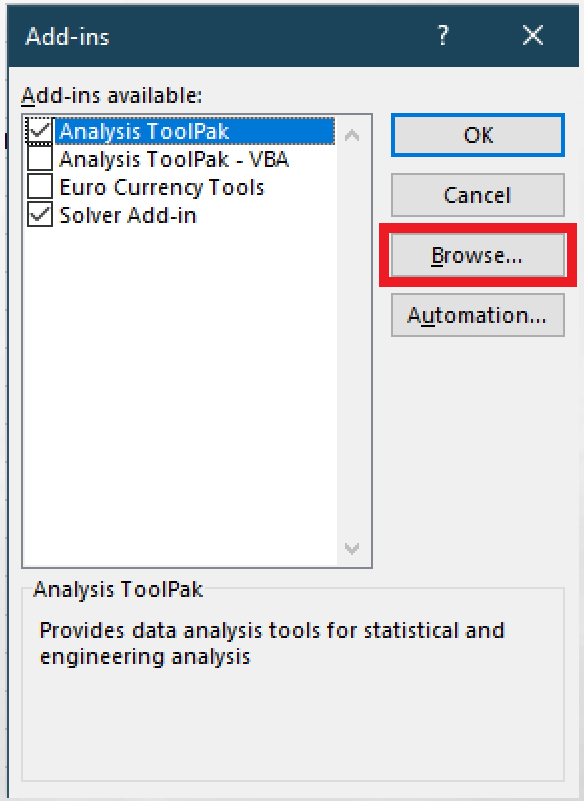
1. Download the Add-in file from
2. Open Excel
3. Go to (File → Option→ Add-in → Go)



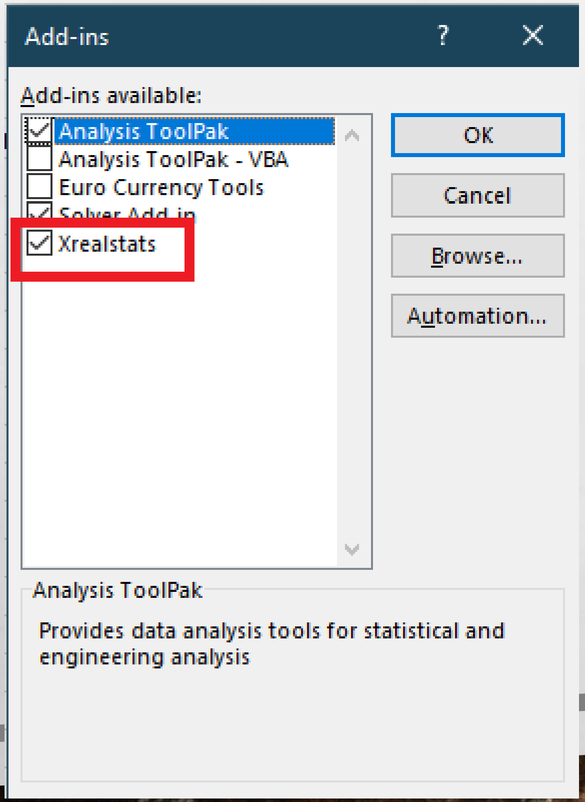
# Steps for Binomial Logistic Regression- Using Real Statistics

**Installing – Real Statistics**

4. Click on Browse and locate the File you downloaded



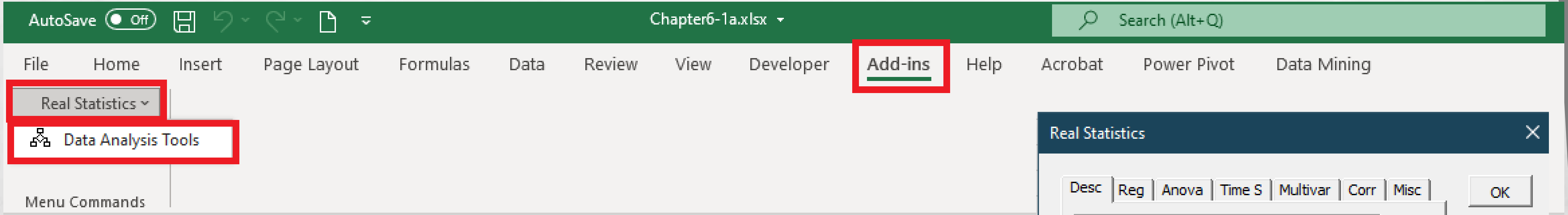

5. Click on the checkbox of (xrealStats)

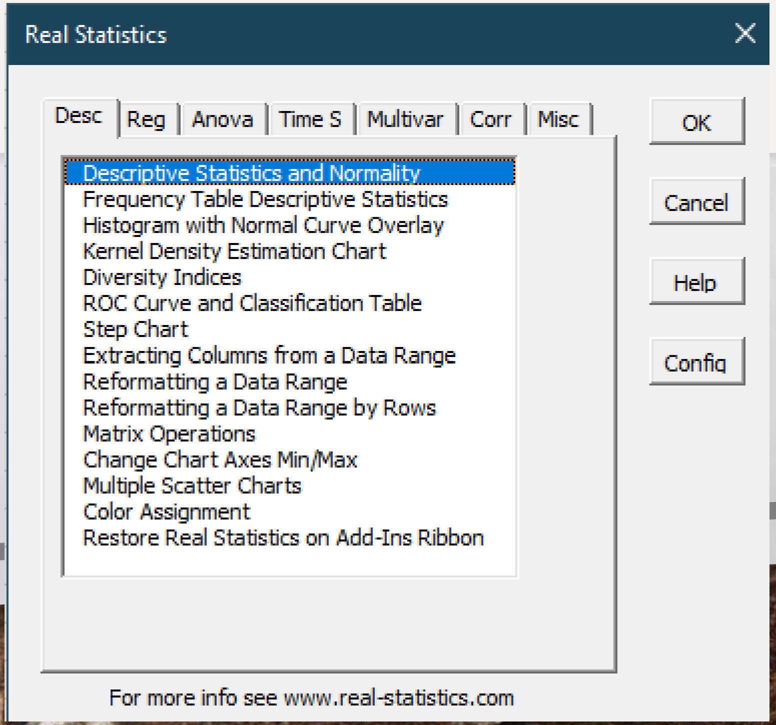


# Steps for Binomial Logistic Regression- Using Real Statistics

**Installing – Real Statistics**

6. Now ,we are ready to use (Real Statistics) Add-in
7. Go to (Add-ins ) Menu → Real Statistics → Data Analysis Tools

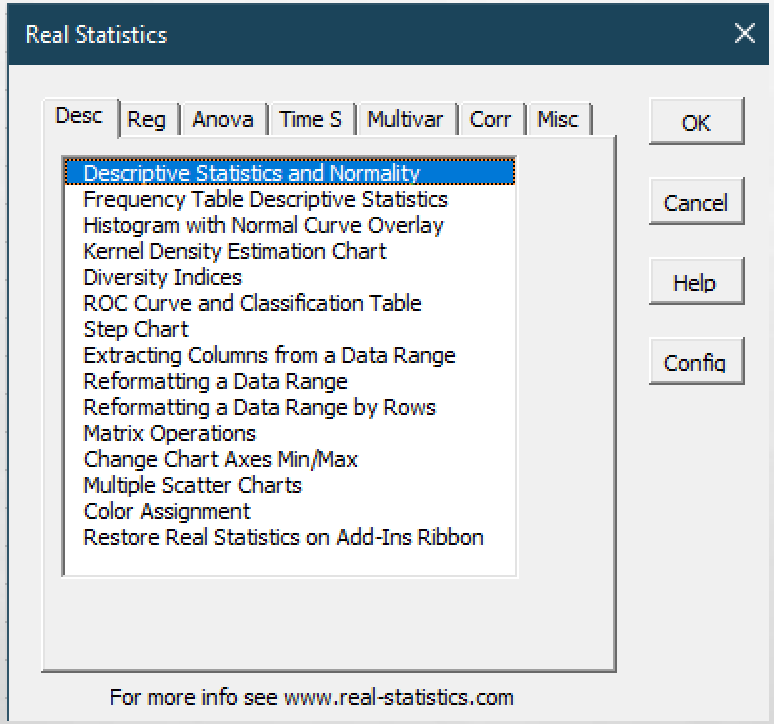


# Steps for Binomial Logistic Regression- Using Real Statistics

**Installing – Real Statistics**

8. From the following interface, we will use the different statistical test to perform logistic regression



# Binomial Logistic Regression:

To perform Binomial Logistic Regression, we will consider the following example that is present through the following (partial) data set.
This data set presents patients that survived cancer in the last five years and the effect of 5 different genes on the survival results that is present in the last column.

**Note: The Excel file related to this example contains 87 records**

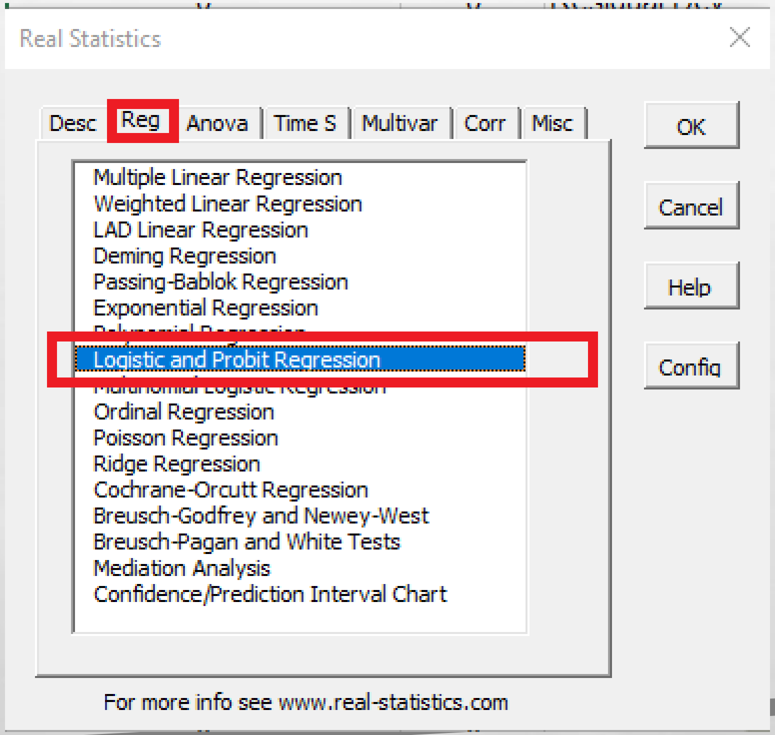
| patientID | gene1 | gene2 | gene3 | gene4 | gene5 | 5-year survival |
|---|---|---|---|---|---|---|
| 1 | 92.0826 | 443.3735 | 350.9466 | 11.1876 | 77.926 | 0 |
| 2 | 97.1228 | 29.21562 | 2.579007 | 301.8684 | 171.9968 | 0 |
| 3 | 7.73995 | 42.36842 | 39.90712 | 9.2879 | 104.2632 | 1 |
| 4 | 60.2125 | 25.63164 | 2.420307 | 14.1677 | 94.6316 | 1 |
| 5 | 385.4766 | 20.32964 | 5.831751 | 35.4298 | 71.0588 | 0 |
| 6 | 476.644 | 12.48745 | 4.406125 | 50.444 | 104.7838 | 0 |
| 7 | 25.89 | 18.49515 | 28.72168 | 6.4724 | 78.3981 | 0 |
| 8 | 142.8572 | 15.41199 | 1.019015 | 24.3784 | 146.7328 | 1 |

# Binomial Logistic Regression:

We can see from the previous data that the column named (5-year Survival) is the prediction column (dependent variable) and the previous columns (gene1…gene5) are the (independent variables) that will will used to predict the result of dependent variable. Here we can use binomial logistic regression as the result of dependent variable are binary (such as (0,1) / (Black ,white) / (True, False)…etc)

# Steps for Solving Binomial Logistic Regression Example

1. Open the Excel File
2. Go to (Add-ins)→ Real Statistics → xrealstats
3. In (xrealstats) choose →Reg Tab → (Logistic and Probit Regression)



# Steps for Solving Binomial Logistic Regression Example

4. Select the following parameters, it is important to know the order of data, so that the dependent variable (Y) is always displayed in the last column as shown in the following figure.

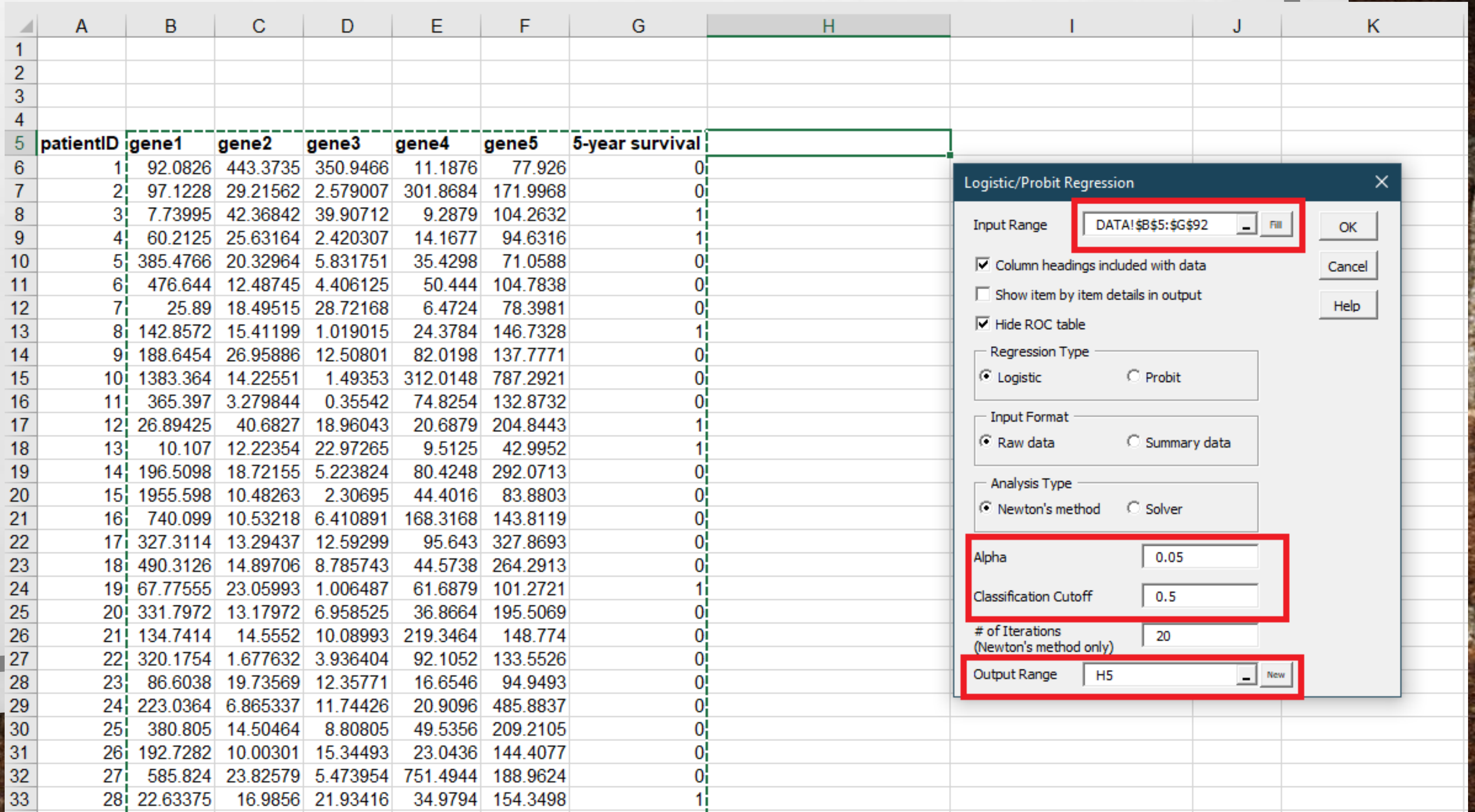
**Moreover, we see that the column headings is included with data**

**No need to have patientID**

**Alpha value is 0.05**

**The cutoff is set to be 0.5**

**The Output Range is (H5)**



| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |
| 5 | patientID | gene1 | gene2 | gene3 | gene4 | gene5 | 5-year survival |
| 6 | 1 | 92.0826 | 443.3735 | 350.9466 | 11.1876 | 77.926 | 0 |
| 7 | 2 | 97.1228 | 29.21562 | 2.579007 | 301.8684 | 171.9968 | 0 |
| 8 | 3 | 7.73995 | 42.36842 | 39.90712 | 9.2879 | 104.2632 | 1 |
| 9 | 4 | 60.2125 | 25.63164 | 2.420307 | 14.1677 | 94.6316 | 1 |
| 10 | 5 | 385.4766 | 20.32964 | 5.831751 | 35.4298 | 71.0588 | 0 |
| 11 | 6 | 476.644 | 12.48745 | 4.406125 | 50.444 | 104.7838 | 0 |
| 12 | 7 | 25.89 | 18.49515 | 28.72168 | 6.4724 | 78.3981 | 0 |
| 13 | 8 | 142.8572 | 15.41199 | 1.019015 | 24.3784 | 146.7328 | 1 |
| 14 | 9 | 188.6454 | 26.95886 | 12.50801 | 82.0198 | 137.7771 | 0 |
| 15 | 10 | 1383.364 | 14.22551 | 1.49353 | 312.0148 | 787.2921 | 0 |
| 16 | 11 | 365.397 | 3.279844 | 0.35542 | 74.8254 | 132.8732 | 0 |
| 17 | 12 | 26.89425 | 40.6827 | 18.96043 | 20.6879 | 204.8443 | 1 |
| 18 | 13 | 10.107 | 12.22354 | 22.97265 | 9.5125 | 42.9952 | 1 |
| 19 | 14 | 196.5098 | 18.72155 | 5.223824 | 80.4248 | 292.0713 | 0 |
| 20 | 15 | 1955.598 | 10.48263 | 2.30695 | 44.4016 | 83.8803 | 0 |
| 21 | 16 | 740.099 | 10.53218 | 6.410891 | 168.3168 | 143.8119 | 0 |
| 22 | 17 | 327.3114 | 13.29437 | 12.59299 | 95.643 | 327.8693 | 0 |
| 23 | 18 | 490.3126 | 14.89706 | 8.785743 | 44.5738 | 264.2913 | 0 |
| 24 | 19 | 67.77555 | 23.05993 | 1.006487 | 61.6879 | 101.2721 | 1 |
| 25 | 20 | 331.7972 | 13.17972 | 6.958525 | 36.8664 | 195.5069 | 0 |
| 26 | 21 | 134.7414 | 14.5552 | 10.08993 | 219.3464 | 148.774 | 0 |
| 27 | 22 | 320.1754 | 1.677632 | 3.936404 | 92.1052 | 133.5526 | 0 |
| 28 | 23 | 86.6038 | 19.73569 | 12.35771 | 16.6546 | 94.9493 | 0 |
| 29 | 24 | 223.0364 | 6.865337 | 11.74426 | 20.9096 | 485.8837 | 0 |
| 30 | 25 | 380.805 | 14.50464 | 8.80805 | 49.5356 | 209.2105 | 0 |
| 31 | 26 | 192.7282 | 10.00301 | 15.34493 | 23.0436 | 144.4077 | 0 |
| 32 | 27 | 585.824 | 23.82579 | 5.473954 | 751.4944 | 188.9624 | 0 |
| 33 | 28 | 22.63375 | 16.9856 | 21.93416 | 34.9794 | 154.3498 | 1 |



# Steps for Solving Binomial Logistic Regression Example

5. You are supposed to have the following output

| Row | gene1 | gene2 | gene3 | gene4 | gene5 | 5-year survival |
|---|---|---|---|---|---|---|
| 6 | 92.0826 | 443.3735 | 350.9466 | 11.1876 | 77.926 | 0 |
| 7 | 97.1228 | 29.21562 | 2.579007 | 301.8684 | 171.9968 | 0 |
| 8 | 7.73995 | 42.36842 | 39.90712 | 9.2879 | 104.2632 | 1 |
| 9 | 60.2125 | 25.63164 | 2.420307 | 14.1677 | 94.6316 | 1 |
| 10 | 385.4766 | 20.32964 | 5.831751 | 35.4298 | 71.0588 | 0 |
| 11 | 476.644 | 12.48745 | 4.406125 | 50.444 | 104.7838 | 0 |
| 12 | 25.89 | 18.49515 | 28.72168 | 6.4724 | 78.3981 | 0 |
| 13 | 142.8572 | 15.41199 | 1.019015 | 24.3784 | 146.7328 | 1 |
| 14 | 188.6454 | 26.95886 | 12.50801 | 82.0198 | 137.7771 | 0 |
| 15 | 1383.364 | 14.22551 | 1.49353 | 312.0148 | 787.2921 | 0 |
| 16 | 365.397 | 3.279844 | 0.35542 | 74.8254 | 132.8732 | 0 |
| 17 | 26.89425 | 40.6827 | 18.96043 | 20.6879 | 204.8443 | 1 |
| 18 | 10.107 | 12.22354 | 22.97265 | 9.5125 | 42.9952 | 1 |
| 19 | 196.5098 | 18.72155 | 5.223824 | 80.4248 | 292.0713 | 0 |
| 20 | 1955.598 | 10.48263 | 2.30695 | 44.4016 | 83.8803 | 0 |
| 21 | 740.099 | 10.53218 | 6.410891 | 168.3168 | 143.8119 | 0 |
| 22 | 327.3114 | 13.29437 | 12.59299 | 95.643 | 327.8693 | 0 |
| 23 | 490.3126 | 14.89706 | 8.785743 | 44.5738 | 264.2913 | 0 |
| 24 | 67.77555 | 23.05993 | 1.006487 | 61.6879 | 101.2721 | 1 |
| 25 | 331.7972 | 13.17972 | 6.958525 | 36.8664 | 195.5069 | 0 |
| 26 | 134.7414 | 14.5552 | 10.08993 | 219.3464 | 148.774 | 0 |
| 27 | 320.1754 | 1.677632 | 3.936404 | 92.1052 | 133.5526 | 0 |
| 28 | 86.6038 | 19.73569 | 12.35771 | 16.6546 | 94.9493 | 0 |
| 29 | 223.0364 | 6.865337 | 11.74426 | 20.9096 | 485.8837 | 0 |
| 30 | 380.805 | 14.50464 | 8.80805 | 49.5356 | 209.2105 | 0 |
| 31 | 192.7282 | 10.00301 | 15.34493 | 23.0436 | 144.4077 | 0 |
| 32 | 585.824 | 23.82579 | 5.473954 | 751.4944 | 188.9624 | 0 |
| 33 | 22.63375 | 16.9856 | 21.93416 | 34.9794 | 154.3498 | 1 |
| 34 | 281.9732 | 7.576127 | 4.445798 | 9.7442 | 62.3374 | 1 |
| 35 | 320.488 | 11.43577 | 3.753606 | 23.8576 | 63.9675 | 0 |
| 36 | 775.3844 | 4.270233 | 3.202675 | 0 | 137.3745 | 0 |
| 37 | 220.5718 | 14.02896 | 19.25733 | 151.5038 | 111.1493 | 0 |
| 38 | 51.4593 | 8.617512 | 9.585253 | 21.5054 | 32.7942 | 1 |
| 39 | 122.3764 | 9.053718 | 6.670224 | 72.572 | 318.4899 | 0 |
| 40 | 69.5498 | 36.72441 | 39.82781 | 47.4204 | 104.0339 | 0 |

Logistic Regression

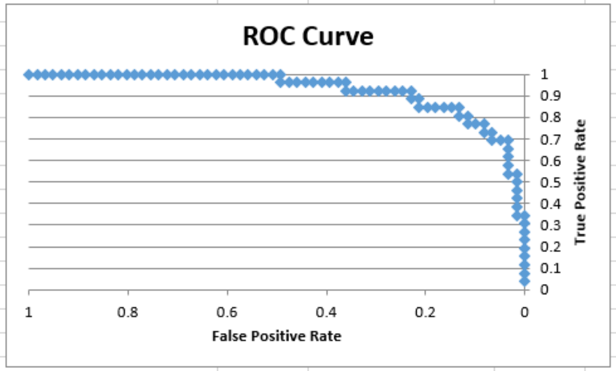
| | | # Iter | 20 | Alpha | | 0.05 | |
|---|---|---|---|---|---|---|---|
| | *coeff* | *s.e.* | *Wald* | *p-value* | *exp(b)* | *lower* | *upper* |
| intercept | 4.22311043 | 1.39318 | 9.188616351 | 0.002435 | 68.24542711 | | |
| gene1 | -0.020320077 | 0.006534 | 9.6716329 | 0.001871 | 0.979884984 | 0.967416 | 0.992514377 |
| gene2 | -0.007939915 | 0.017808 | 0.198785752 | 0.655703 | 0.992091523 | 0.958061 | 1.027330584 |
| gene3 | -0.097140223 | 0.042013 | 5.345949152 | 0.02077 | 0.907428755 | 0.835701 | 0.985313229 |
| gene4 | -0.010731814 | 0.007324 | 2.146794742 | 0.142868 | 0.989325567 | 0.975225 | 1.003630513 |
| gene5 | -0.00379953 | 0.005443 | 0.487317753 | 0.485126 | 0.996207679 | 0.985637 | 1.006891835 |

LL statistics

| | |
|---|---|
| LL | -28.7211 |
| LL0 | -53.0602 |
| Chi-sq | 48.6782 |
| df | 5 |
| p-value | 2.58E-09 |
| R-sq (L) | 0.458707 |
| R-sq (CS) | 0.428516 |
| R-sq (N) | 0.608081 |
| AIC | 69.44218 |
| BIC | 84.23763 |

Covarian

| |
|---|
| 1.94095 |
| -0.0062 |
| -0.012 |
| -0.0397 |
| -0.0016 |
| -0.0029 |



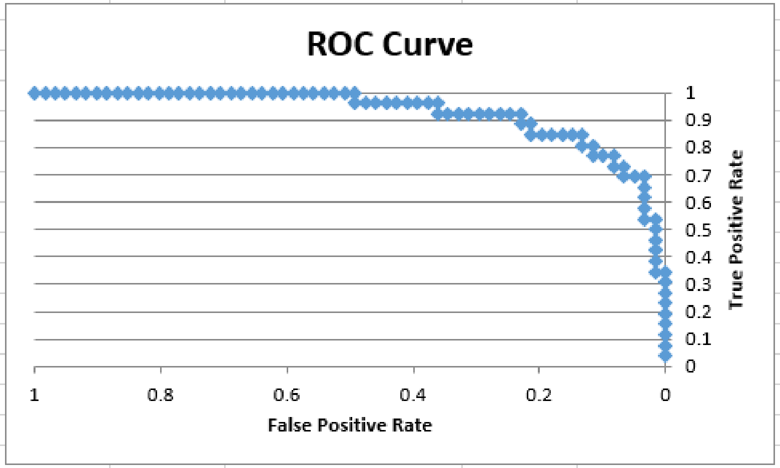
# Steps for Solving Binomial Logistic Regression Example

6. What is important to read in those results

Logistic Regression

# Iter 20 Alpha 0.05

| | coeff | s.e. | Wald | p-value | exp(b) | lower | upper |
|---|---|---|---|---|---|---|---|
| intercept | 4.22311043 | 1.39318 | 9.188616351 | 0.002435 | 68.24542711 | | |
| gene1 | -0.020320077 | 0.006534 | 9.6716329 | 0.001871 | 0.979884984 | 0.967416 | 0.992514377 |
| gene2 | -0.007939915 | 0.017808 | 0.198785752 | 0.655703 | 0.992091523 | 0.958061 | 1.027330584 |
| gene3 | -0.097140223 | 0.042013 | 5.345949152 | 0.02077 | 0.907428755 | 0.835701 | 0.985313229 |
| gene4 | -0.010731814 | 0.007324 | 2.146794742 | 0.142868 | 0.989325567 | 0.975225 | 1.003630513 |
| gene5 | -0.00379953 | 0.005443 | 0.487317753 | 0.485126 | 0.996207679 | 0.985637 | 1.006891835 |



# Steps for Solving Binomial Logistic Regression Example

7. What is the formula for logistic regression

## Binary Logistic Regression

Coefficients

| Term | Coef | SE Coef | 95% CI | Z-Value | P-Value | VIF |
|---|---|---|---|---|---|---|
| Constant | **-9.346** | 0.637 | ( -10.594, -8.097) | -14.67 | 0.000 | |
| FICOscore | **0.014634** | 0.000940 | (0.012791, 0.016476) | 15.56 | 0.000 | 1.00 |

$$\hat{p} = \frac{e^{\beta_0+\beta_1 x_1}}{1+e^{\beta_0+\beta_1 x_1}}$$

Estimated Regression Equation

$$\hat{p} = \frac{e^{-9.346+0.014634x_1}}{1+e^{-9.346+0.014634x_1}}$$

Estimated Regression Equation

# Steps for Solving Binomial Logistic Regression Example

8. What is the formula for our example of logistic regression

$$P = \frac{e^{4.223+\ -0.0203*gene1+\ -0.0079*gene2+\ -0.0971*gene3+\ -0.0107*gene4+\ -0.0037*gene5}}{1 + e^{4.223+\ -0.0203*gene1+\ -0.0079*gene2+\ -0.0971*gene3+\ -0.0107*gene4+\ -0.0037*gene5}}$$

# Steps for Solving Binomial Logistic Regression Example

9. Applying the formula on the dataset (logistic Regression Formula Results)

| 4 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 5 | patientID | gene1 | gene2 | gene3 | gene4 | gene5 | 5-year survival | Logistic Regression Formula Results |
| 6 | 1 | 92.0826 | 443.3735 | 350.9466 | 11.1876 | 77.926 | 0 | 0 |
| 7 | 2 | 97.1228 | 29.21562 | 2.579007 | 301.8684 | 171.9968 | 0 | 0 |
| 8 | 3 | 7.73995 | 42.36842 | 39.90712 | 9.2879 | 104.2632 | 1 | 0 |
| 9 | 4 | 60.2125 | 25.63164 | 2.420307 | 14.1677 | 94.6316 | 1 | 1 |
| 10 | 5 | 385.4766 | 20.32964 | 5.831751 | 35.4298 | 71.0588 | 0 | 0 |
| 11 | 6 | 476.644 | 12.48745 | 4.406125 | 50.444 | 104.7838 | 0 | 0 |
| 12 | 7 | 25.89 | 18.49515 | 28.72168 | 6.4724 | 78.3981 | 0 | 1 |
| 13 | 8 | 142.8572 | 15.41199 | 1.019015 | 24.3784 | 146.7328 | 1 | 1 |
| 14 | 9 | 188.6454 | 26.95886 | 12.50801 | 82.0198 | 137.7771 | 0 | 0 |
| 15 | 10 | 1383.364 | 14.22551 | 1.49353 | 312.0148 | 787.2921 | 0 | 0 |
| 16 | 11 | 365.397 | 3.279844 | 0.35542 | 74.8254 | 132.8732 | 0 | 0 |
| 17 | 12 | 26.89425 | 40.6827 | 18.96043 | 20.6879 | 204.8443 | 1 | 1 |
| 18 | 13 | 10.107 | 12.22354 | 22.97265 | 9.5125 | 42.9952 | 1 | 1 |
| 19 | 14 | 196.5098 | 18.72155 | 5.223824 | 80.4248 | 292.0713 | 0 | 0 |
| 20 | 15 | 1955.598 | 10.48263 | 2.30695 | 44.4016 | 83.8803 | 0 | 0 |
| 21 | 16 | 740.099 | 10.53218 | 6.410891 | 168.3168 | 143.8119 | 0 | 0 |
| 22 | 17 | 327.3114 | 13.29437 | 12.59299 | 95.643 | 327.8693 | 0 | 0 |
| 23 | 18 | 490.3126 | 14.89706 | 8.785743 | 44.5738 | 264.2913 | 0 | 0 |
| 24 | 19 | 67.77555 | 23.05993 | 1.006487 | 61.6879 | 101.2721 | 1 | 1 |
| 25 | 20 | 331.7972 | 13.17972 | 6.958525 | 36.8664 | 195.5069 | 0 | 0 |
| 26 | 21 | 134.7414 | 14.5552 | 10.08993 | 219.3464 | 148.774 | 0 | 0 |
| 27 | 22 | 320.1754 | 1.677632 | 3.936404 | 92.1052 | 133.5526 | 0 | 0 |
| 28 | 23 | 86.6038 | 19.73569 | 12.35771 | 16.6546 | 94.9493 | 0 | 1 |
| 29 | 24 | 223.0364 | 6.865337 | 11.74426 | 20.9096 | 485.8837 | 0 | 0 |
| 30 | 25 | 380.805 | 14.50464 | 8.80805 | 49.5356 | 209.2105 | 0 | 0 |
| 31 | 26 | 192.7282 | 10.00301 | 15.34493 | 23.0436 | 144.4077 | 0 | 0 |
| 32 | 27 | 585.824 | 23.82579 | 5.473954 | 751.4944 | 188.9624 | 0 | 0 |
| 33 | 28 | 22.63375 | 16.9856 | 21.93416 | 34.9794 | 154.3498 | 1 | 1 |
| 34 | 29 | 281.9732 | 7.576127 | 4.445798 | 9.7442 | 62.3374 | 1 | 0 |
| 35 | 30 | 320.488 | 11.43577 | 3.753606 | 23.8576 | 63.9675 | 0 | 0 |
| 36 | 31 | 775.3844 | 4.270233 | 3.202675 | 0 | 137.3745 | 0 | 0 |
| 37 | 32 | 220.5718 | 14.02896 | 19.25733 | 151.5038 | 111.1493 | 0 | 0 |
| 38 | 33 | 51.4593 | 8.617512 | 9.585253 | 21.5054 | 32.7942 | 1 | 1 |
| 39 | 34 | 122.3764 | 9.053718 | 6.670224 | 72.572 | 318.4899 | 0 | 0 |
| 40 | 35 | 69.5498 | 36.72441 | 39.82781 | 47.4204 | 104.0339 | 0 | 0 |

Wrong Predictions

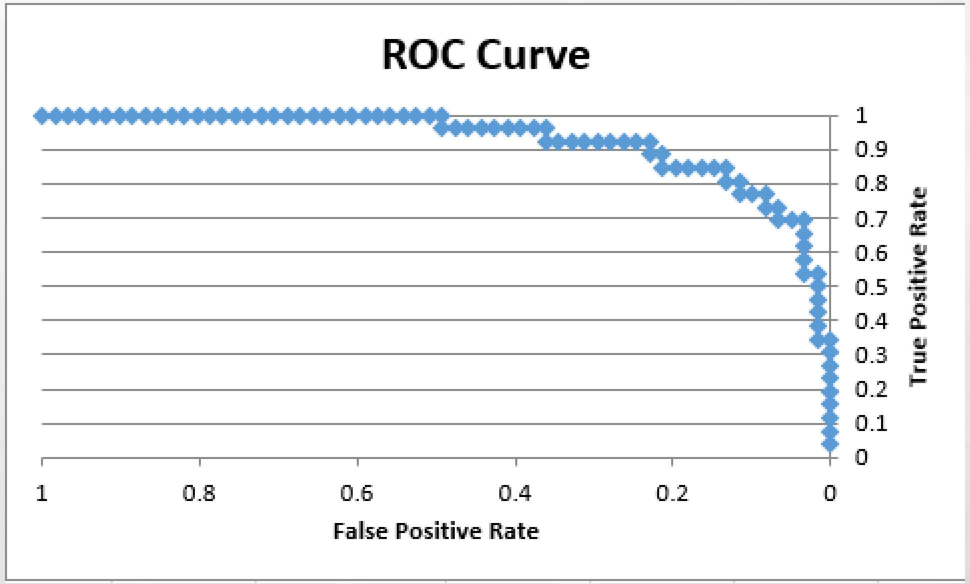
# Steps for Solving Binomial Logistic Regression Example

## 10. Meaning of other results (Classification Table)

Classification Table

| | Obs Suc | Obs Fail | Total |
|---|---|---|---|
| Pred Suc | 21 | 7 | 28 |
| Pred Fail | 5 | 54 | 59 |
| Total | 26 | 61 | 87 |
| Accuracy | 0.807692 | 0.885246 | 0.862069 |

| | |
|---|---|
| Cutoff | 0.5 |
| AUC | 0.92686 |




| | | Actual | |
|---|---|---|---|
| | | Positive | Negative |
| Predicted | Positive | True Positive | False Positive |
| | Negative | False Negative | True Negative |

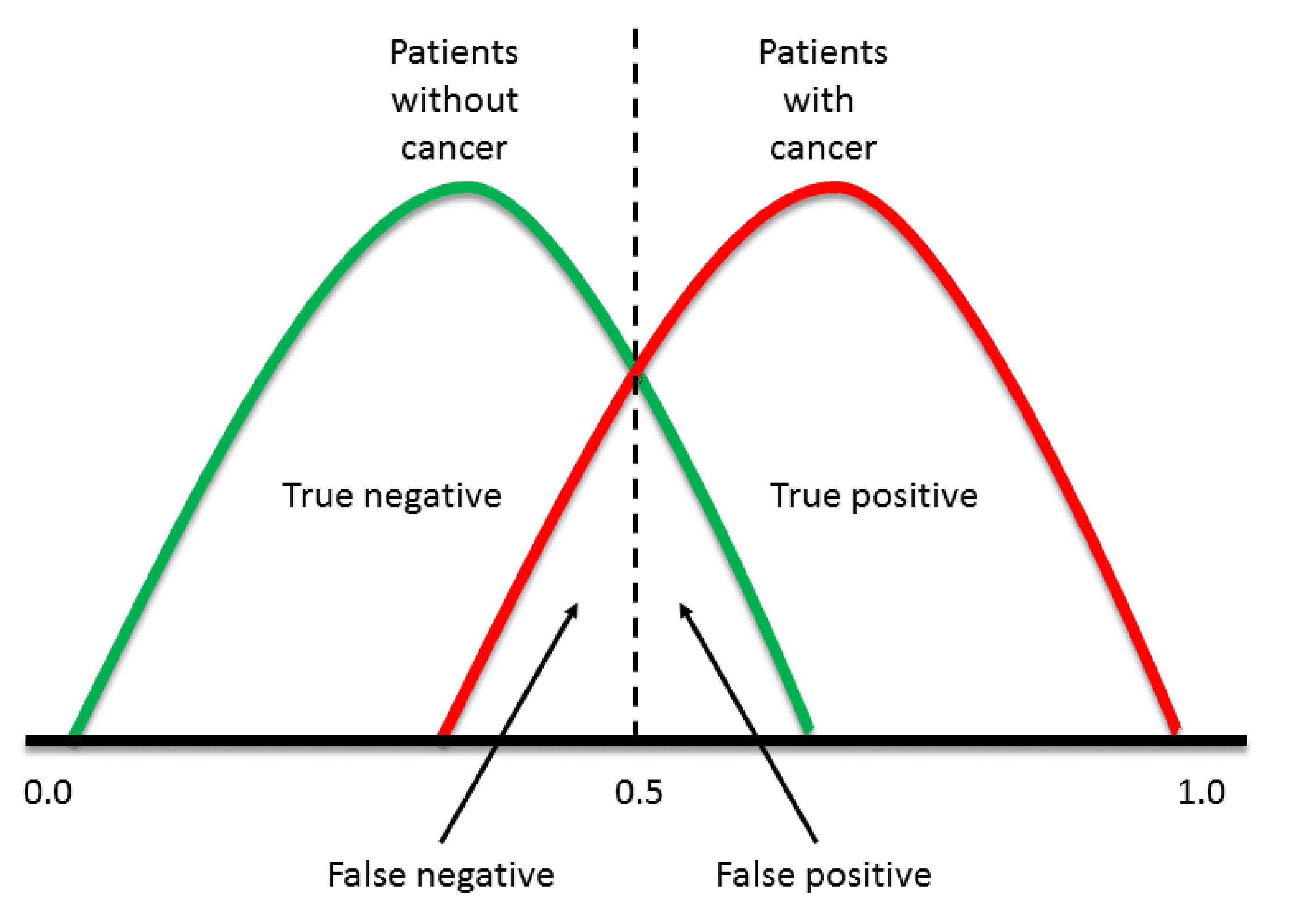
# Steps for Solving Binomial Logistic Regression Example

11. Meaning of other results (Classification Table)

| | | Actual | |
|---|---|---|---|
| | | Positive | Negative |
| Predicted | Positive | **True Positive**<br>*Predicted has cancer*<br>*Has Cancer* | **False Positive**<br>*Predicted has cancer/Does not have cancer* |
| | Negative | **False Negative**<br>*Predicted not cancer*<br>*Has cancer* | **True Negative**<br>*Predicted not cancer*<br>*Does not have cancer* |

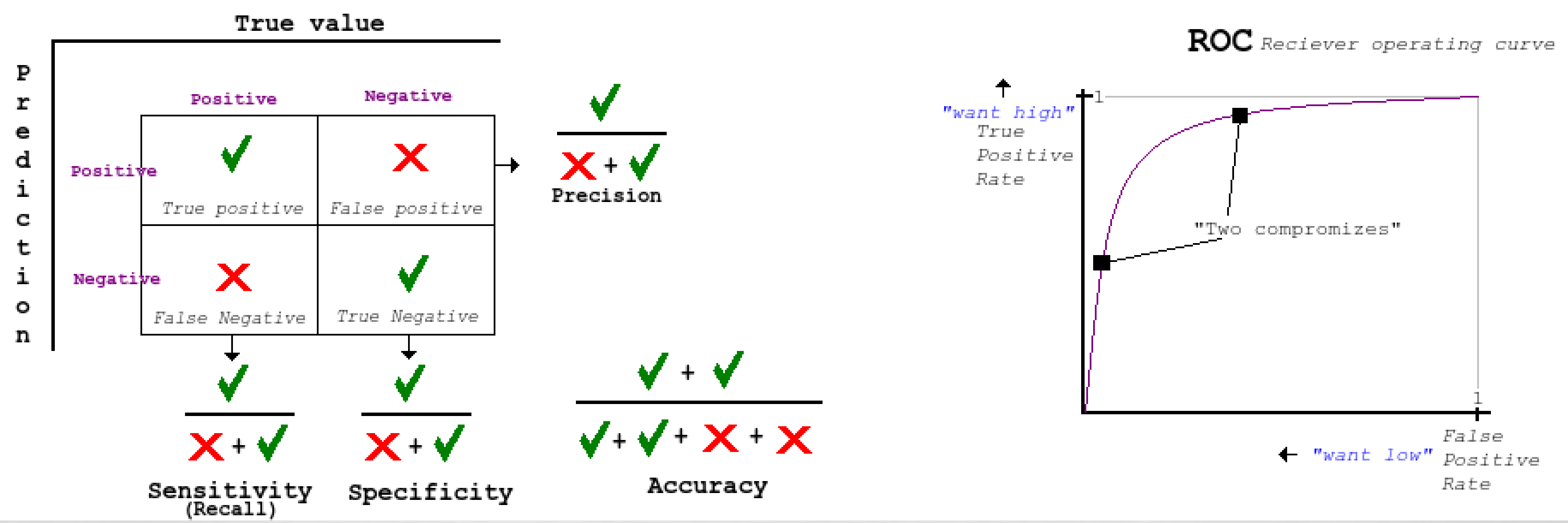
# Steps for Solving Binomial Logistic Regression Example

12. Meaning of other results (Classification Table)



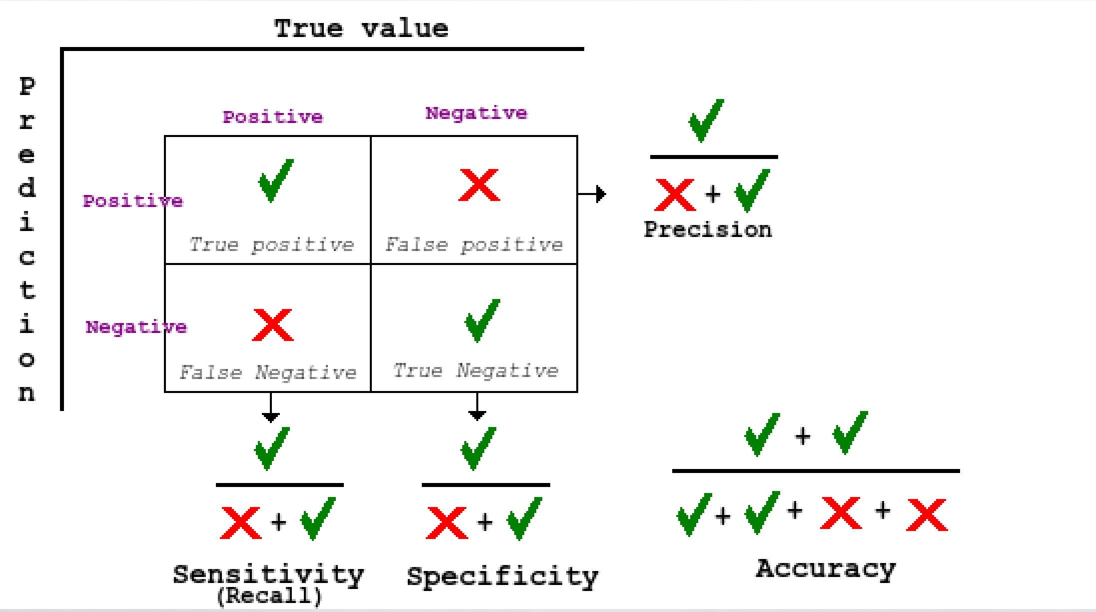
# Steps for Solving Binomial Logistic Regression Example

13. Meaning of other results (Classification Table)



# Steps for Solving Binomial Logistic Regression Example

14. Meaning of other results (Classification Table)




| AB | AC | AD | AE | AF |
|---|---|---|---|---|
| | Classification Table | | | |
| | | Obs Suc | Obs Fail | Total |
| | Pred Suc | 21 | 7 | 28 |
| | Pred Fail | 5 | 54 | 59 |
| | Total | 26 | 61 | 87 |
| | Accuracy | 0.807692 | 0.885246 | 0.862069 |
| | Cutoff | 0.5 | | |
| | AUC | 0.92686 | | |

# Steps for Solving Binomial Logistic Regression Example

15. Meaning of other results (Classification Table)

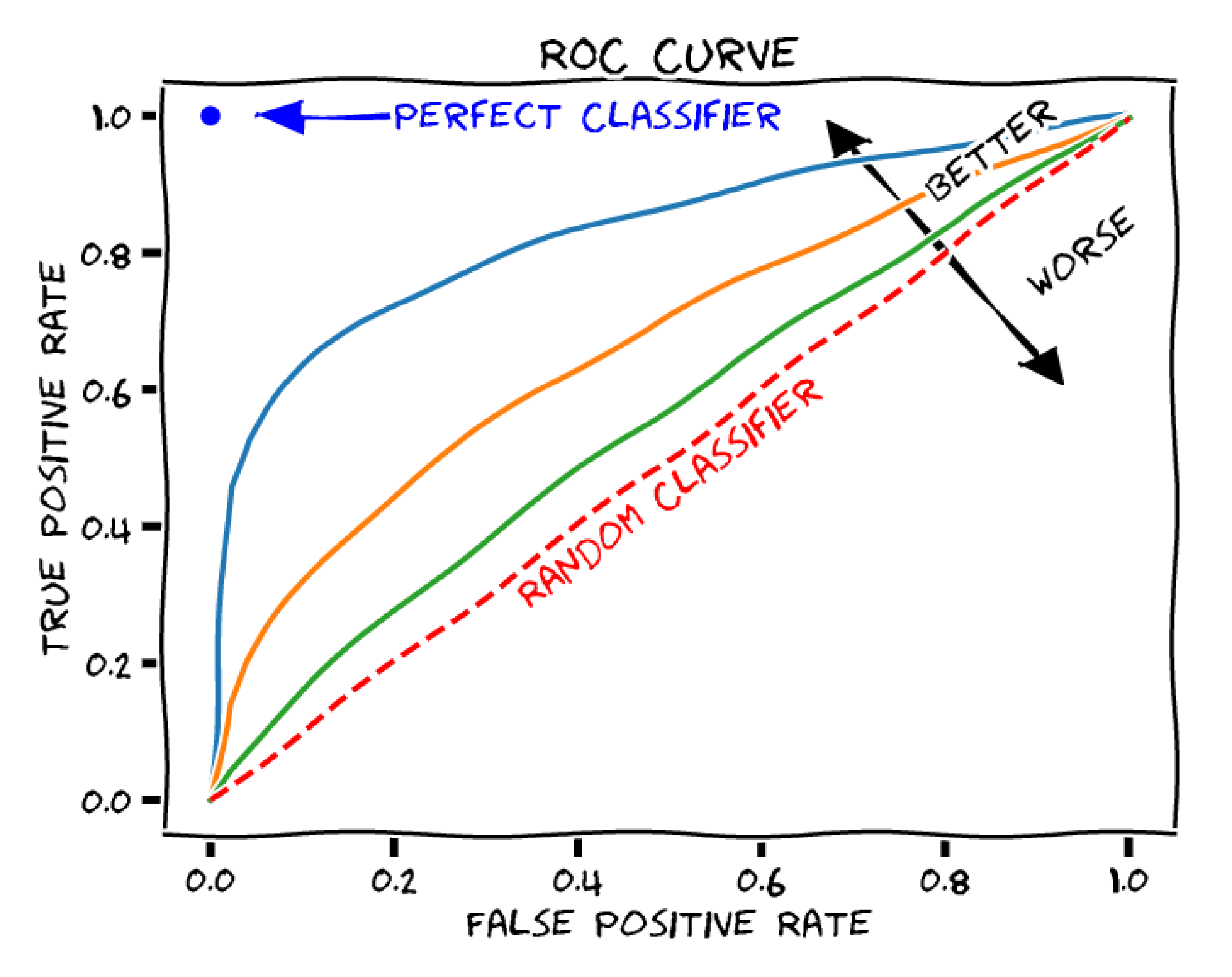
| | | True Health Condition | |
|---|---|---|---|
| | | Has disease | Healthy |
| Diagnosis | Has disease | True positive | False positive |
| | Healthy | False negative | True negative |

**Sensitivity** = True positive / Has disease

**Specificity** = True negative / Healthy

# Steps for Solving Binomial Logistic Regression Example

16. Meaning of other results (Classification Table)



# Steps for Solving Binomial Logistic Regression Example

17. How to enhance the predictions

a. Using the cut-off value (increase or decrease)

**b. Watch the coefficient values >0.05 and eliminate the variables and make the test and predictions again.**

# Steps for Solving Binomial Logistic Regression Example

17. How to enhance the predictions

a. Using the cut-off value **(increase or decrease)**

Classification Table

| | Obs Suc | Obs Fail | Total |
|---|---|---|---|
| Pred Suc | 21 | 7 | 28 |
| Pred Fail | 5 | 54 | 59 |
| Total | 26 | 61 | 87 |
| Accuracy | 0.807692 | 0.885246 | 0.862069 |
| Cutoff | 0.5 | | |
| AUC | 0.92686 | | |

Classification Table

| | Obs Suc | Obs Fail | Total |
|---|---|---|---|
| Pred Suc | 22 | 10 | 32 |
| Pred Fail | 4 | 51 | 55 |
| Total | 26 | 61 | 87 |
| Accuracy | 0.846154 | 0.836066 | 0.83908 |
| Cutoff | 0.4 | | |
| AUC | 0.92686 | | |

Classification Table

| | Obs Suc | Obs Fail | Total |
|---|---|---|---|
| Pred Suc | 18 | 2 | 20 |
| Pred Fail | 8 | 59 | 67 |
| Total | 26 | 61 | 87 |
| Accuracy | 0.692308 | 0.967213 | 0.885057 |
| Cutoff | 0.6 | | |
| AUC | 0.92686 | | |

**Cutoff = 0.5**

**Cutoff = 0.4**

**Cutoff = 0.6**

# Steps for Solving Binomial Logistic Regression Example

17. How to enhance the predictions

**b. Watch the coefficient values >0.05 and eliminate the variables and make the test and predictions again.**

| Logistic Regression | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | # Iter | 20 | | Alpha | 0.05 | |
| | *coeff* | *s.e.* | *Wald* | *p-value* | *exp(b)* | *lower* | *upper* |
| intercept | 4.22311043 | 1.39318 | 9.188616351 | 0.002435 | 68.24542711 | | |
| gene1 | -0.020320077 | 0.006534 | 9.6716329 | 0.001871 | 0.979884984 | 0.967416 | 0.992514377 |
| gene2 | -0.007939915 | 0.017808 | 0.198785752 | 0.655703 | 0.992091523 | 0.958061 | 1.027330584 |
| gene3 | -0.097140223 | 0.042013 | 5.345949152 | 0.02077 | 0.907428755 | 0.835701 | 0.985313229 |
| gene4 | -0.010731814 | 0.007324 | 2.146794742 | 0.142868 | 0.989325567 | 0.975225 | 1.003630513 |
| gene5 | -0.00379953 | 0.005443 | 0.487317753 | 0.485126 | 0.996207679 | 0.985637 | 1.006891835 |

The variables highlighted in blue are showing less effect on the prediction as their p-values is larger than 0.05, Thus we can eliminate those variables and recalculate the prediction and modify the prediction model.

# Multinomial Logistic Regression:

The Multinomial Logistic Regression is a case where the dependent variables are more than 2 (Binomial) , they can be 3 or more (Multinomial). To understand this case, we will have the following example.

In this example we have a dataset that contains information about employees, and the preferred insurance product each employee had (dependent variable). Moreover, we have 5 different independent variables that are (age, household, position-level , gender, absent).

The data in the dataset needs to be modified to be numerical instead of categorial or nominal.

Thus, we will have the values of (0 for A , 1 for B, 2 for C), we will have the values of (1 for Male, 2 for Female).

Moreover, we will move the (product) to be the last column

**Note: The Excel file related to this example contains 1454 records**

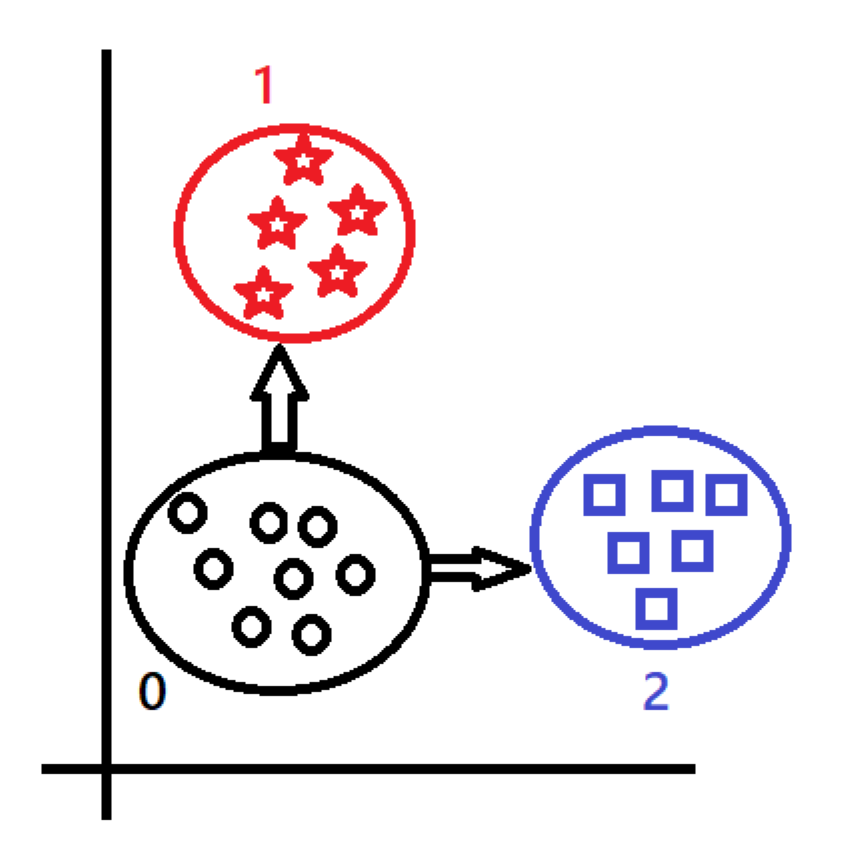
| product | age | household | position_level | gender | absent |
|---|---|---|---|---|---|
| C | 57 | 2 | 2 | Male | 10 |
| A | 21 | 7 | 2 | Male | 7 |
| C | 66 | 7 | 2 | Male | 1 |
| A | 36 | 4 | 2 | Female | 6 |
| A | 23 | 0 | 2 | Male | 11 |
| A | 31 | 5 | 1 | Male | 14 |
| A | 37 | 3 | 3 | Male | 12 |
| B | 37 | 0 | 3 | Female | 25 |
| C | 55 | 3 | 3 | Female | 3 |
| B | 66 | 2 | 4 | Female | 18 |
| B | 58 | 1 | 2 | Female | 1 |
| C | 62 | 2 | 2 | Female | 25 |
| A | 31 | 2 | 2 | Female | 0 |

# Multinomial Logistic Regression:

| age | household | position_level | gender | absent | product |
|---|---|---|---|---|---|
| 57 | 2 | 2 | 1 | 10 | 2 |
| 21 | 7 | 2 | 1 | 7 | 0 |
| 66 | 7 | 2 | 1 | 1 | 2 |
| 36 | 4 | 2 | 2 | 6 | 0 |
| 23 | 0 | 2 | 1 | 11 | 0 |
| 31 | 5 | 1 | 1 | 14 | 0 |
| 37 | 3 | 3 | 1 | 12 | 0 |
| 37 | 0 | 3 | 2 | 25 | 1 |
| 55 | 3 | 3 | 2 | 3 | 2 |
| 66 | 2 | 4 | 2 | 18 | 1 |
| 58 | 1 | 2 | 2 | 1 | 1 |
| 62 | 2 | 2 | 2 | 25 | 2 |
| 31 | 2 | 2 | 2 | 0 | 0 |
| 45 | 5 | 2 | 1 | 10 | 2 |
| 63 | 3 | 4 | 1 | 20 | 2 |

# Multinomial Logistic Regression:

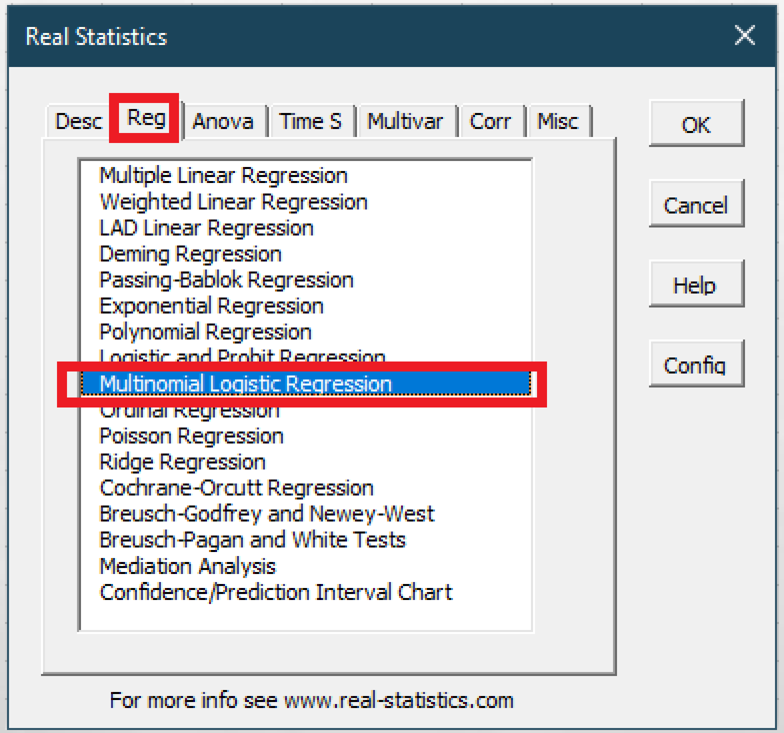
The type of Multinomial Logistic Regression is (One Vs One).
We have to set one of insurance products as main , we will have the 0 as main one
**Then we will compare 0 → 1 / and 0 → 2**



# Multinomial Logistic Regression:

**Steps to perform Multinomial Logistic Regression.**

1. Choose (Add-ins) Menu → Real Statistics → Data Analysis Tools
2. On the (reg) tab choose (Multinomial Logistic Regression)



# Multinomial Logistic Regression:

**Steps to perform Multinomial Logistic Regression.**

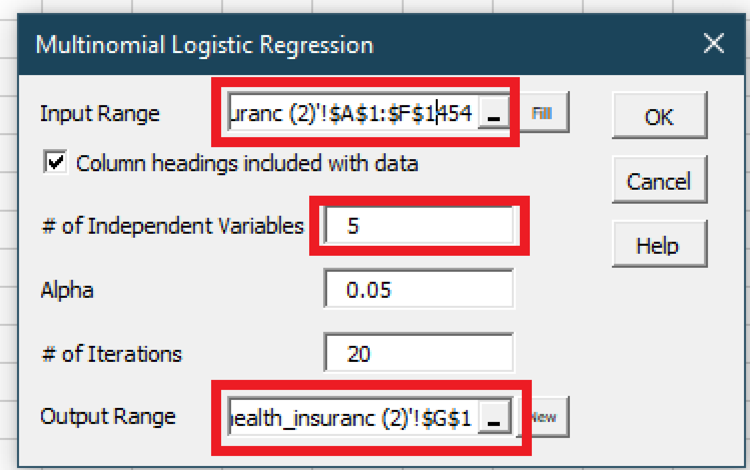
**3. Select the (input Range) to hold all the data, and make sure that the (dependent variable) is the last column**

**4. Define number of independent variables (In this example they are 5)**

**5. Define the (Output Range) in this example (G1)**

**6. Click on OK**

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | age | household | position_level | gender | absent | product |
| 2 | 21 | 0 | 2 | 1 | 13 | 0 |
| 3 | 21 | 0 | 3 | 1 | 9 | 0 |
| 4 | 21 | 0 | 3 | 1 | 17 | 0 |
| 5 | 21 | 0 | 5 | 1 | 24 | 0 |
| 6 | 21 | 0 | 5 | 2 | 20 | 0 |
| 7 | 21 | 0 | 5 | 2 | 30 | 0 |
| 8 | 21 | 1 | 1 | 2 | 23 | 0 |
| 9 | 21 | 1 | 2 | 1 | 10 | 0 |
| 10 | 21 | 1 | 2 | 2 | 0 | 0 |
| 11 | 21 | 1 | 2 | 2 | 11 | 0 |
| 12 | 21 | 1 | 3 | 2 | 8 | 0 |
| 13 | 21 | 1 | 3 | 2 | 21 | 0 |
| 14 | 21 | 1 | 5 | 2 | 20 | 0 |
| 15 | 21 | 2 | 2 | 1 | 25 | 0 |



# Multinomial Logistic Regression:

**Steps to perform Multinomial Logistic Regression.**
**7. If all steps are performed correctly , you will have the following result:**

**Multinomial Logistic Regression - Data Summary**

| age | household | position_level | gender | absent | 0 | 1 | 2 |
|---|---|---|---|---|---|---|---|
| 21 | 0 | 2 | 1 | 13 | 1 | 0 | 0 |
| 21 | 0.00 | 3 | 1 | 9 | 1 | 0 | 0 |
| 21 | 0 | 3 | 1 | 17 | 1 | 0 | 0 |
| 21 | 0 | 5 | 1 | 24 | 1 | 0 | 0 |
| 21 | 0 | 5 | 2 | 20 | 1 | 0 | 0 |
| 21 | 0 | 5 | 2 | 30 | 1 | 0 | 0 |
| 21 | 1 | 1 | 2 | 23 | 1 | 0 | 0 |
| 21 | 1 | 2 | 1 | 10 | 1 | 0 | 0 |
| 21 | 1 | 2 | 2 | 0 | 1 | 0 | 0 |
| 21 | 1 | 2 | 2 | 11 | 1 | 0 | 0 |
| 21 | 1 | 3 | 2 | 8 | 1 | 0 | 0 |
| 21 | 1 | 3 | 2 | 21 | 1 | 0 | 0 |
| 21 | 1 | 5 | 2 | 20 | 1 | 0 | 0 |
| 21 | 2 | 2 | 1 | 25 | 1 | 0 | 0 |
| 21 | 2 | 4 | 1 | 12 | 1 | 0 | 0 |
| 21 | 2 | 4 | 1 | 25 | 1 | 0 | 0 |
| 21 | 2 | 5 | 2 | 24 | 1 | 0 | 0 |
| 21 | 3 | 2 | 2 | 16 | 1 | 0 | 0 |
| 21 | 3 | 3 | 2 | 25 | 1 | 0 | 0 |
| 21 | 3 | 4 | 1 | 28 | 1 | 0 | 0 |
| 21 | 3 | 4 | 2 | 18 | 1 | 0 | 0 |
| 21 | 4 | 3 | 1 | 0 | 1 | 0 | 0 |
| 21 | 4 | 3 | 1 | 2 | 1 | 0 | 0 |
| 21 | 4 | 3 | 1 | 11 | 1 | 0 | 0 |
| 21 | 5 | 1 | 2 | 13 | 1 | 0 | 0 |
| 21 | 5 | 2 | 1 | 10 | 1 | 0 | 0 |
| 21 | 5 | 2 | 2 | 3 | 1 | 0 | 0 |
| 21 | 5 | 4 | 1 | 13 | 1 | 0 | 0 |

Multinomial Logistic Regression

Alpha 0.05

Reference Variable 0

1

| | coeff | s.e. | Wald | p-value | xp(b) | lower | upper |
|---|---|---|---|---|---|---|---|
| intercept | -9.171016192 | 0.667 | 188.8 | 0 | E-04 | | |
| age | 0.242042461 | 0.015 | 248.5 | 0 | 1.274 | 1.236 | 1.313 |
| household | -0.958495528 | 0.069 | 193.8 | 0 | 0.383 | 0.335 | 0.439 |
| position_leve | -0.41132374 | 0.089 | 21.51 | 3.51305E-06 | 0.663 | 0.557 | 0.789 |
| gender | 2.302130635 | 0.229 | 101 | 0 | 9.995 | 6.38 | 15.66 |
| absent | 0.010261795 | 0.013 | 0.632 | 0.426675988 | 1.01 | 0.985 | 1.036 |

2

| | coeff | s.e. | Wald | p-value | xp(b) | lower | upper |
|---|---|---|---|---|---|---|---|
| intercept | -9.978409994 | .673 | 219.9 | 0 | 5E-05 | | |
| age | 0.269096724 | .016 | 297.1 | 0 | 1.309 | 1.269 | 1.349 |
| household | 0.204827065 | 0.05 | 17.05 | 3.63776E-05 | 1.227 | 1.114 | 1.353 |
| position_leve | -0.213146219 | .082 | 6.735 | 0.009456126 | 0.808 | 0.688 | 0.949 |
| gender | -0.119660149 | .195 | 0.378 | 0.538850878 | 0.887 | 0.606 | 1.299 |
| absent | 0.003123222 | .012 | 0.064 | 0.800902238 | 1.003 | 0.979 | 1.028 |

R-square

| | |
|---|---|
| LL | -748 |
| LL0 | -1595 |
| R-sq (L) | 0.531 |
| R-sq (CS) | 0.688 |
| R-sq (N) | 0.774 |
| AIC | 1521 |
| BIC | 1585 |

Significance Test

| | |
|---|---|
| LL | -748 |
| LL0 | -1595 |
| Chi-sq | 1693 |
| df | 10 |
| p-value | 0 |
| # iter | 20 |

# Multinomial Logistic Regression:

**Steps to perform Multinomial Logistic Regression.**

**8. We wrote two formulas for predicting (0 -1) and (0-2), and those formulas are considered our prediction model as it is shown in the following figure.**

G2 =EXP($L$6+$L$7*A2+$L$8*B2+$L$9*C2+$L$10*D2+$L$11*E2)/(1+EXP($L$6+$L$7*A2+$L$8*B2+$L$9*C2+$L$10*D2+$L$11*E2))

| | age | household | position_le | gender | absent | product | Regression Formula 0 - 1 | | Regression Formula 0 - 2 |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 21 | 0 | 2 | 1 | 13 | 0 | 0.077612421 | 0 | 0.007901816 |
| 3 | 21 | 0 | 3 | 1 | 9 | 0 | 0.050805508 | | 0.006315778 |
| 4 | 21 | 0 | 3 | 1 | 17 | 0 | 0.054913638 | | 0.006474536 |
| 5 | 21 | 0 | 5 | 1 | 24 | 0 | 0.026692187 | | 0.004330154 |
| 6 | 21 | 0 | 5 | 2 | 20 | 0 | 0.208292932 | | 0.003796145 |
| 7 | 21 | 0 | 5 | 2 | 30 | 0 | 0.225721779 | | 0.003916107 |
| 8 | 21 | 1 | 1 | 2 | 23 | 0 | 0.350313556 | | 0.010952397 |
| 9 | 21 | 1 | 2 | 1 | 10 | 0 | 0.030338811 | | 0.009591197 |
| 10 | 21 | 1 | 2 | 2 | 0 | 0 | 0.220113298 | | 0.008258951 |
| 11 | 21 | 1 | 2 | 2 | 11 | 0 | 0.240100883 | | 0.008545154 |
| 12 | 21 | 1 | 3 | 2 | 8 | 0 | 0.168788808 | | 0.006852097 |
| 13 | 21 | 1 | 3 | 2 | 21 | 0 | 0.188340061 | | 0.007134004 |
| 14 | 21 | 1 | 5 | 2 | 20 | 0 | 0.09164261 | | 0.004655041 |
| 15 | 21 | 2 | 2 | 1 | 25 | 0 | 0.013801402 | | 0.012302224 |
| 16 | 21 | 2 | 4 | 1 | 12 | 0 | 0.00535083 | | 0.007748395 |
| 17 | 21 | 2 | 4 | 1 | 25 | 0 | 0.006109786 | | 0.008066878 |
| 18 | 21 | 2 | 5 | 2 | 24 | 0 | 0.038746791 | | 0.005778484 |
| 19 | 21 | 3 | 2 | 2 | 16 | 0 | 0.046627794 | | 0.013015174 |
| 20 | 21 | 3 | 3 | 2 | 25 | 0 | 0.034330875 | | 0.010840398 |
| 21 | 21 | 3 | 4 | 1 | 28 | 0 | 0.002425123 | | 0.009974564 |

It is in favour of product 0 from both comparision , thus we put the prediction value as 0

| Multinomial Logistic Regression | | | | | Alpha | 0.05 | |
|---|---|---|---|---|---|---|---|
| Reference Variable | 0 | | | | | | |
| 1 | | | | | | | |
| | coeff | s.e. | Wald | p-value | exp(b) | lower | upper |
| intercept | -9.171016192 | 0.667413 | 188.818823 | 0 | 0.000104011 | | |
| age | 0.242042461 | 0.015354 | 248.519599 | 0 | 1.273848281 | 1.236086 | 1.3 |
| household | -0.958495528 | 0.068846 | 193.832764 | 0 | 0.383469371 | 0.335065 | 0. |
| position_level | -0.41132374 | 0.08868 | 21.5137272 | 3.51E-06 | 0.662772331 | 0.557032 | 0.7 |
| gender | 2.302130635 | 0.229066 | 101.004152 | 0 | 9.995456455 | 6.380011 | 15. |
| absent | 0.010261795 | 0.01291 | 0.63185422 | 0.426675988 | 1.010314628 | 0.985072 | 1.0 |
| 2 | | | | | | | |
| | coeff | s.e. | Wald | p-value | exp(b) | lower | upper |
| intercept | -9.978409994 | 0.672891 | 219.904077 | 0 | 4.64E-05 | | |
| age | 0.269096724 | 0.015612 | 297.080308 | 0 | 1.308781725 | 1.26934 | 1.3 |
| household | 0.204827065 | 0.049603 | 17.0516037 | 3.64E-05 | 1.227312802 | 1.113611 | 1. |
| position_level | -0.213146219 | 0.082134 | 6.7345844 | 0.009456126 | 0.808037978 | 0.68789 | 0.9 |
| gender | -0.119660149 | 0.194711 | 0.37767368 | 0.538850878 | 0.887221909 | 0.60575 | 1.2 |
| absent | 0.003123222 | 0.012385 | 0.06359456 | 0.800902238 | 1.003128105 | 0.979071 | 1.0 |

# Multinomial Logistic Regression:

**9. In this slide we have found the accuracy of this model by counting the actual occurrences of each chosen product and compare it with the results of the predicted values.**
**The sum of Actual values is divided with the predicted one which gives the (Accuracy) of the model**

| | | | | | | |
|---|---|---|---|---|---|---|
| | | **Predicted** | | | | |
| **Actual** | | | 0 | 1 | 2 | **Sum** |
| | 0 | 495 | 489 | 453 | 499 | **1441** |
| | 1 | 459 | | | | |
| | 2 | 499 | | **Accuracy** | | **0.992** |
| **Sum** | | **1453** | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | **Predicted** | | | | |
| **Actual** | | | 0 | 1 | 2 | **Sum** |
| | 0 | COUNTIF(F9:F1461,"=0") | COUNTIF(L4:L1450,">0") | COUNTIF(M9:M1450,">0") | COUNTIF(N9:N1450,">0") | **1441** |
| | 1 | COUNTIF(F9:F1461,"=1") | | | | |
| | 2 | COUNTIF(F9:F1461,"=2") | | **Accuracy** | | **0.992** |
| **Sum** | | **1453** | | | | |

# Multinomial Logistic Regression:

**10. To answer the question if we can enhance our model, we need to look at P-value for both tables:**
**We can see that in both tables (Absent) is the mutual independent variable that can be removed as it does not have main effect on the dependent variable. Thus by removing it, it will enhance our prediction model.**
**Note: When removed from data , you need to perform the multinomial logistic regression again and have new results.**

| K | L | M | N | O | P | Q | R |
|---|---|---|---|---|---|---|---|
| **Multinomial Logistic Regression** | | | | | **Alpha** | **0.05** | |
| **Reference Variable** | | **0** | | | | | |
| | | | | | | | |
| 1 | | | | | | | |
| | **coeff** | **s.e.** | **Wald** | **p-value** | **exp(b)** | **lower** | **upper** |
| **intercept** | -9.171016192 | 0.667413 | 188.818823 | 0 | 0.00010401 | | |
| **age** | 0.242042461 | 0.015354 | 248.519599 | 0 | 1.27384828 | 1.236086 | 1.312764258 |
| **household** | -0.958495528 | 0.068846 | 193.832764 | 0 | 0.38346937 | 0.335065 | 0.43886627 |
| **position_level** | -0.41132374 | 0.08868 | 21.5137272 | 3.51E-06 | 0.66277233 | 0.557032 | 0.788585828 |
| **gender** | 2.302130635 | 0.229066 | 101.004152 | 0 | 9.99545646 | 6.380011 | 15.65971444 |
| **absent** | 0.010261795 | 0.01291 | 0.63185422 | 0.42667599 | 1.01031463 | 0.985072 | 1.036204243 |
| | | | | | | | |
| **2** | | | | | | | |
| | **coeff** | **s.e.** | **Wald** | **p-value** | **exp(b)** | **lower** | **upper** |
| **intercept** | -9.978409994 | 0.672891 | 219.904077 | 0 | 4.64E-05 | | |
| **age** | 0.269096724 | 0.015612 | 297.080308 | 0 | 1.30878173 | 1.26934 | 1.349449313 |
| **household** | 0.204827065 | 0.049603 | 17.0516037 | 3.64E-05 | 1.2273128 | 1.113611 | 1.35262404 |
| **position_level** | -0.213146219 | 0.082134 | 6.7345844 | 0.00945613 | 0.80803798 | 0.68789 | 0.949170521 |
| **gender** | -0.119660149 | 0.194711 | 0.37767368 | 0.53885088 | 0.88722191 | 0.60575 | 1.299483665 |
| **absent** | 0.003123222 | 0.012385 | 0.06359456 | 0.80090224 | 1.00312811 | 0.979071 | 1.027775958 |